PROJECT REPORT 2

GUESSING THE MATHEMATICAL RANDOM NUMBER

PROJECT OVERVIEW

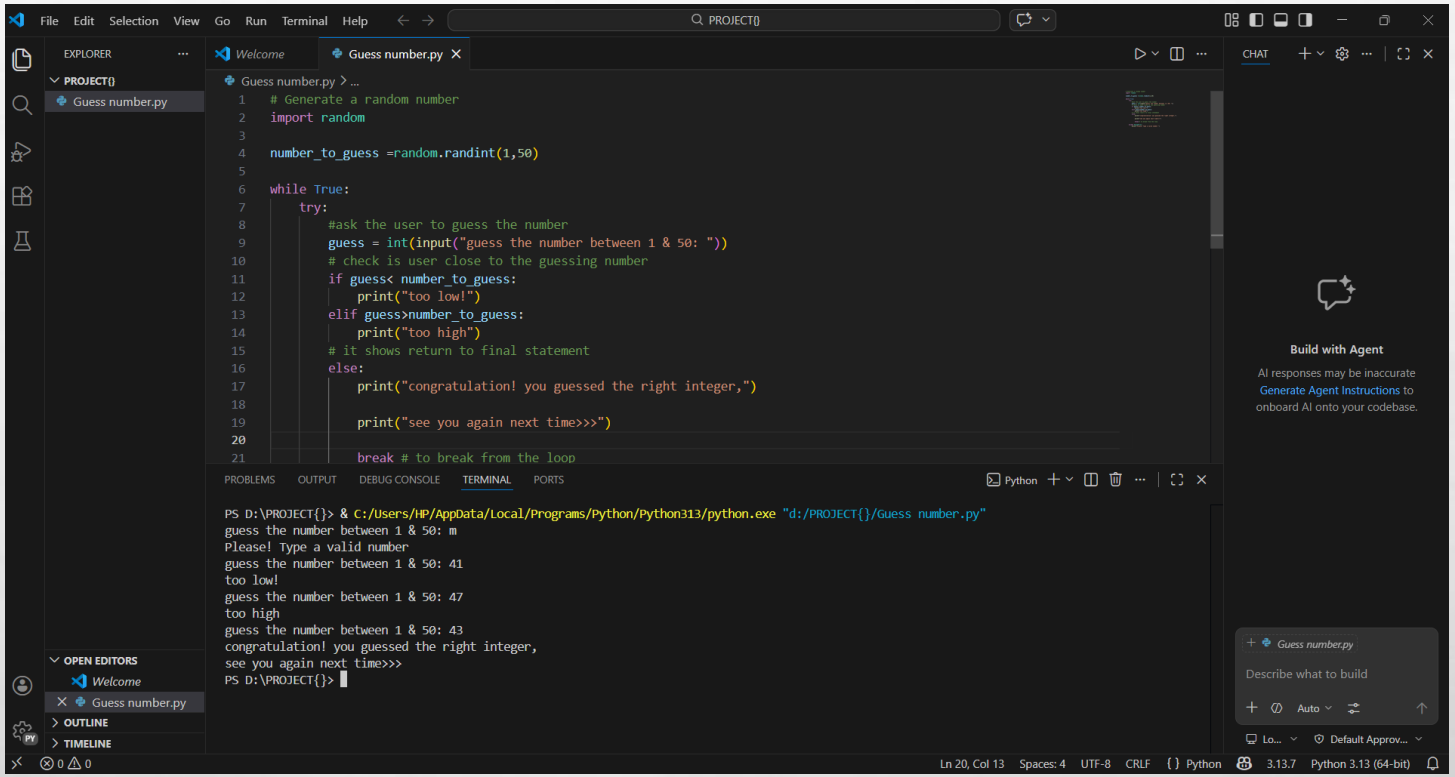
THE OBJECTIVE OF THIS PROJECT IS TO DESIGN A SYSTEM WHERE A COMPUTER GENERATE A PSEUDO-RANDOM INTEGER WITHIN A SPECIFIED RANGE (I.E., 1 TO 50), AND A USER ATTEMPTS TO GUESS IT. THE SYSTEM PROVIDES FEEDBACK- TOO HIGH OR TOO LOW – UNTIL THE CORRECT NUMBER IS FOUND.

- THE MATHEMATICAL FOUNDATION

WHILE THE COMPUTER CHOOSE A NUMBER RANDOMLY, THE USER'S STRATEGY SHOULD BE ANYTHIN BUT RANDOM. THE MOST EFFICIENT WAY TO SOLVE THIS IS THROUGH THE BINARY SEARCH ALGORITHM.

BY ALWAYS GUESSING THE MIDPOINT OF THE CURRENT RANGE, YOU ELIMINATE HALF OF THE REMAINING POSSIBILITIES WITH EVERY SINGLE GUESS.

IMAGE OF THE PROGRAM USED IN PROJECT



LIBRARY USED IN THIS PROJECT

- IMPORT RANDOM : THIS LIBRARY USED BY THE DEVICE TO GENERATE A RANDOM NUMBER IN PROGRAM
- RANDINT : WHILE USING THIS FUNCTION IT GIVES TO ALLOW THE USE OF INT FUNCTION IN ALGORITHM

THANKING YOU